
Avani Gupta

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Education

IIIT Hyderabad

Hyderabad, India

B.Tech (Honours) + Masters By Research in Computer Science & Engineering, 2023

Advisor: Prof. P. J. Narayanan, MS specialization: AI, MS CGPA: 9/10

Thesis: Leveraging Human-Centered Explanations for Model Improvement & Evaluation [Link](#)

Skills

Languages/Libraries: Python, C/C++, PyTorch, TensorFlow, JAX, transformers.

Frameworks: Polars, RLLib, MARL (Mujoco, Gym), Distributed Training (DDP, FSDP, DeepSpeed), Synthetic Dataset Generation (Distilabel), AI Agents Development (MCP, A2A protocols, LangChain, ADK, Open-Claw), FastAPI, SQL.

Research Experience

MBZUAI Client AI Engineer and Researcher

Abu Dhabi, UAE

Jun 2025 - Present

- Translated industry problems into research formulations, facilitating faculty-industry partnerships which led to 5+ active collaborations and signed MOUs with stakeholders.
- Built AI Agents for various internal and stakeholder use-cases using k2-think series of foundation model released by IFM.
- Owned evaluation and safety checks of RAG assistant: created scalable pipelines and generated synthetic persona-conditioned datasets, evaluated for correctness and safety.
- Built AI Agentic prototypes for email assistant and department level team intelligence for team tasks management from emails and meetings.
- Independently proposed research on collective-survival-based multi-agent reinforcement learning paradigm aimed to induce pro-social values and inculcate explicit learning from both success and failures of peers; introduced novel peer-learning loss & survival-shaped rewards with promising results on benchmarks like SISL and Smac-v2 (research ongoing with [Prof. Abdalla Swikir](#) and [Prof. Imran Razzak](#)).

Stealth AI Engineer

Abu Dhabi, UAE

March 2024 - May 2025

- Architected and owned end-to-end development of production-scale agents serving 3,000+ users (multi-modal RAG, SQL, voice-call, marketing content generation including social media posts and high quality posters) using MCP/A2A protocols.
- Also built several components including custom safety stack having content moderation, jailbreak and toxicity detection and flagging, & hallucination filters.
- Trained small LLMs (1-3B parameters, 2T tokens) for efficient edge deployment (using FSDP, megatron); proposed a novel MLA+GQA hybrid attention mechanism improving compute efficiency for coding & tool-calling capabilities.
- Owned post-training small LLMs for code and tool calling: generated synthetic datasets and did supervised finetuning and evaluation which achieved 2-3% improvements on code/tool calling benchmarks.

G42 Healthcare AI Research Associate

Abu Dhabi, UAE

Mar 2023 - Mar 2024

- Owned EHR foundation model development: modeled large-scale EHR sequences & trained a foundation model for patient trajectory prediction (processed 2.2 million records to get 4.3B tokens dataset), including chronic disease predictions, personalized medicine recommendation & mortality forecasting.
- Obtained patient embeddings revealing clinical coherence (chronic disease clusters, comorbidities) and EHR foundation model achieving high scores (0.99 precision and 0.99 recall in top@5) in prediction of next procedures, conditions (including chronic diseases), and medications.
- Curated training corpora from 10M+ articles & contributed to evaluation pipelines for [Med42](#) (70B open-

source medical LLM achieving 72% on USMLE, AAAI FM Symposium 2024) and 89B dataset for building trust and bias analysis (EMNLP 2025).

Centre for Visual Information Technology (CVIT), IIIT Hyderabad Researcher

Hyderabad, India

May 2020 - Mar 2023

- Originated a unified line of research under my advisor [Prof. P. J. Narayanan](#) on concept-based learning for inducing knowledge priors, interpretable control, and debiasing neural networks.
- Introduced the concept-sensitivity loss (NeurIPS 23) & sensitivity-based disentanglement metric (ICVGIP 22).
- Guided students in research on using activation space manipulations for backdoor defense (ICCV 25) & controllable style transfer (ICVGIP 25).
- Also worked on neural rendering, particularly NeRFs and temporal consistency in loose clothing animations of 3D Human with [Prof. Avinash Sharma](#).

IBM Research Research Intern

Bangalore, India

Sep 2022 - Dec 2022

- Researched on anomalous IT events prediction using transformers and published my work in CODS-COMAD 2024.

IBM Research Research Intern

Bangalore, India

May 2021 - Aug 2021

- Formulated reinforcement learning based next-best-action recommendation for business processes with conflicting goals; trained PPO with action masking to handle dynamically evolving action spaces from process flow graphs (deployed on IBM Business Automation Workflow, US patent application pending).

Honors & Awards

- **Best Paper Award**, ICVGIP 2022 1/400+ papers. [\[Link\]](#)
- **Global Rank 14**, Amazon ML Challenge 2021 3,000+ teams, 12K participants. [\[Link\]](#)
- **National Winner**, Smart India Hackathon 2020 Top 0.1% from 100,000+ teams. [\[Link\]](#)
- **National Winner**, Microsoft Mars Program 2020 Selected from 100,000+ students. [\[Link\]](#)
- **Dean's List** (Top 1% SGPA, Monsoon 2021) & **Merit List** (Top 5% SGPA, Spring 2021), IIIT Hyderabad
- **Merit-Based Chief Minister's State Scholarship** (2017, 2018), Chhattisgarh

Service & Leadership

- **Reviewer**: ICML 2026, CVPR 2026, NeurIPS 2025, ACL 2024, AAAI 2023.
- **Teaching & Mentorship**: Teaching Assistant for *Statistical Methods in AI* (Spring 2022); gave guest talks on NeRFs, GANs, and interpretability at the 3D Vision Summer School and IIIT-H Computer Graphics course (2021 – 2022); AI Tutor and Mentor for the TalentSprint Professional AI Program (2022 – 2023).
- **Workshops & Community Engagement**: Instructor for the GANs Workshop [\[GitHub\]](#); Organizer of the AICrowd *ML Battleground* challenge (2021) [\[Link\]](#); Student Volunteer at NeurIPS RL Workshop (2021).
- **Student Organizations**: Events Wing Member, Robotics Club, IIIT Hyderabad.

Peer-Reviewed Publications

- **Prototype Guided Backdoor Defense via Activation Space Manipulation**, ICCV 2025. Venkat Adithya Amula, Sunayana Samavedam, Saurabh Saini, **Avani Gupta**[†], P. J. Narayanan. [\[paper\]](#) | [\[project page\]](#)
- **CAV Styler: Interpretable & Controllable Style Transfer**, ICVGIP 2025. Sunayana Samavedam, Venkat Adithya Amula, Saurabh Saini, **Avani Gupta**[†], P. J. Narayanan.
- **Concept Distillation: Leveraging Human-Centered Explanations for Model Improvement** NeurIPS 2023. **Avani Gupta**, Saurabh Saini, P. J. Narayanan. [\[paper\]](#) | [\[project page\]](#)
- **Interpreting Intrinsic Image Decomposition using Concept Activations**, ICVGIP 2022 (Oral + Best Paper). **Avani Gupta**, Saurabh Saini, P. J. Narayanan. [\[paper\]](#) | [\[project page\]](#)
- **Building Trust in Clinical LLMs: Bias Analysis & Dataset Transparency**, EMNLP 2025.

Svetlana Maslenkova, Clement Christophe, Marco AF Pimentel, Tathagata Raha, Muhammad Umar Salman, Ahmed Al Mahrooqi, **Avani Gupta**, Shadab Khan, Ronnie Rajan, Praveenkumar Kanithi. [\[paper\]](#)

- **Med42 - Evaluating Fine-Tuning Strategies for Medical LLMs: Full-Parameter vs. Parameter-Efficient Approaches**, AAAI FM Symposium 2024.
Clement Christophe, Praveenkumar Kanithi, Prateek Munjal, Tathagata Raha, Nasir Hayat, Ronnie Rajan, Ahmed Al Mahrooqi, **Avani Gupta**, Muhammad Umar Salman, Marco AF Pimentel, Shadab Khan, Boulbaba Ben Amor. [\[paper\]](#) [\[HuggingFace\]](#)
- **Predicting Business Process Events in Presence of Anomalous IT Events**, CODS-COMAD 2024. **Avani Gupta**, Avirup Saha, Sambit Ghosh, Neelamadhav Gantayat, Renuka Sindhgatta. [\[paper\]](#)
- **CitRet: A Hybrid Model for Cited Text Span Retrieval**, COLING 2022.
Amit Pandey*, **Avani Gupta***, Vikram Pudi. [\[paper\]](#).

Preprints

- **A survey on Concept-based Approaches For Model Improvement**, **Avani Gupta**, P. J. Narayanan. [arXiv:2403-14566](#)
- **Goal-Oriented Next Best Activity Recommendation using Reinforcement Learning**, Prerna Agarwal*, **Avani Gupta***, Renuka Sindhgatta, Sampath Dechu. [arXiv:2305.03219](#)

Patent

- **Generating organizational goal-oriented & process-conformant recommendation models using artificial intelligence techniques**, [\[US Patent Application\]](#), Status: Pending. Prerna Agarwal, Sampath Dechu, **Avani Gupta**, Renuka Sindhgatta

References

Prof. P. J. Narayanan Professor, IIIT Hyderabad (Masters Advisor) pnj@iiit.ac.in	Renuka Sindhgatta STSM, IBM Research (Former Research Manager) renuka.sindhgatta.rajan@ibm.com	Marco Pimentel Associate Director - AI & Applied Sciences, M42 (Former Research Manager) marco.and.pimentel@gmail.com
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